

This exam has 6 pages and 7 exercises.

The result will be computed as (total number of points plus 10) divided by 10.

Please motivate all answers!

1. Associativity (9 points)

Is bi-implication $\leftrightarrow$ associative? If so, prove this via a truth table. If not, give a counterexample.

Solution: Bi-implication is associative. The argumentation via a truth table is:

ϕ	ψ	χ	$\phi \leftrightarrow \psi$	$\psi \leftrightarrow \chi$	$(\phi \leftrightarrow \psi) \leftrightarrow \chi$	$\phi \leftrightarrow (\psi \leftrightarrow \chi)$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	F	F	F	F
T	F	F	F	T	T	T
F	T	T	F	T	F	F
F	T	F	F	F	T	T
F	F	T	T	F	T	T
F	F	F	T	T	F	F

2. Sheffer stroke (12 points)

Express the propositional formula $\neg(\phi \oplus \psi)$ using only the Sheffer stroke $|$.

(Hint: Use the semantic equivalence $\phi \oplus \psi \equiv \neg(\phi \wedge \psi) \wedge \neg(\neg\phi \wedge \neg\psi)$.)

$$\begin{aligned}
 \neg(\phi \oplus \psi) &\equiv \neg(\neg(\phi \wedge \psi) \wedge \neg(\neg\phi \wedge \neg\psi)) \\
 &\equiv \neg((\phi|\psi) \wedge \neg((\phi|\phi) \wedge (\psi|\psi))) \\
 &\equiv \neg((\phi|\psi) \wedge ((\phi|\phi)|(\psi|\psi))) \\
 &\equiv (\phi|\psi)|((\phi|\phi)|(\psi|\psi))
 \end{aligned}$$

3. Island puzzle (12 points)

On the island of liars and truth speakers, everybody is either a liar (who always lies) or a truth speaker (who always speaks the truth). You meet islanders A , B , and C .

A says: “Either B or C is a truth speaker.” (This is an exclusive or, i.e., $\oplus$.)

B says: “ A and C are both liars.”

Determine, either through logical reasoning or via a truth table, who of A , B , and C is a truth speaker and who is a liar.

Solution: Logical reasoning approach: Suppose B is a truth speaker.

Then, by B 's statement, A and C are both liars.

But then A 's statement is true (since B is a truth speaker and C a liar), contradicting that A is a liar.

So we conclude that B is a liar.

Suppose A is a liar. Then A 's statement is false, so since B is a liar, C must also be a liar.

But then B 's claim is true, contradicting that B is a liar.

So we conclude that A is a truth speaker.

Then, by A 's statement, only one of B and C is a liar. Since B is a liar, **we conclude that C is a truth speaker.**

Truth table approach:

t_A	t_B	t_C	$t_B \oplus t_C$	$\neg t_A \wedge \neg t_C$	$t_A \leftrightarrow (t_B \oplus t_C)$	$t_B \leftrightarrow (\neg t_A \wedge \neg t_C)$
T	T	T	F	F	F	F
T	T	F	T	F	T	F
T	F	T	T	F	T	T
T	F	F	F	F	F	T
F	T	T	F	F	T	F
F	T	F	T	T	F	T
F	F	T	T	F	F	T
F	F	F	F	T	T	F

Only in the 3rd row do the 6th and 7th column both have a T. So A and C are truth speakers and B is a liar.

4. DPLL procedure (12 points)

Apply the DPLL procedure to the CNF $(p \vee q \vee r) \wedge (p \vee \neg q \vee \neg r) \wedge \neg p$, to check whether it is satisfiable. Explain in detail how the procedure comes to a result.

Solution: In the first step, since $\neg p$ is a unit clause, $\perp$ is substituted for it.

In 4 reductions, the formula reduces to $(q \vee r) \wedge (\neg q \vee \neg r)$.

In the second step, one can pick q or r and substitute $\top$ for it.

If one picks q , in 4 reductions the formula reduces to $\neg r$.

Substituting $\top$ for r yields $\perp$.

Next, substituting $\perp$ for r yields $\top$.

So then the found solution is p is $\perp$, q is $\top$, and r is $\perp$.

If one picks r , in 4 reductions the formula reduces to $\neg q$.

Substituting $\top$ for q yields $\perp$.

Next, substituting $\perp$ for q yields $\top$.

So then the found solution is p is $\perp$, q is $\perp$, and r is $\top$.

5. Sets (6 + 9 points)

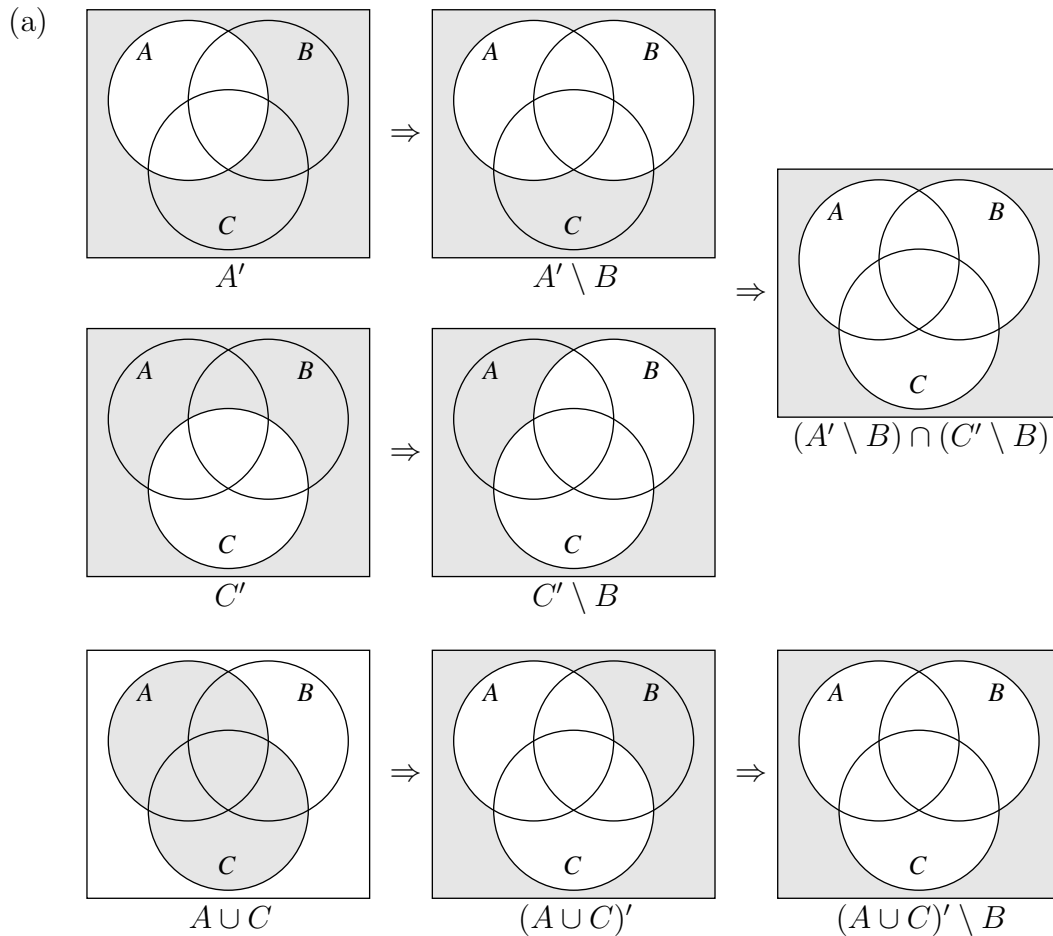
Consider the following set-theoretic equality

$$(A' \setminus B) \cap (C' \setminus B) = (A \cup C)' \setminus B.$$

(a) Using the method you have learned in the lectures, for each side of the equality, draw a sequence of Venn diagrams that clearly illustrates how the set is constructed from A , B and C using the fundamental set operations, and use this to show that the set-theoretic equality holds for all sets A , B and C .

(b) Prove the set-theoretic equality using the rules of the algebra of sets.

Solution:



(b)

$$\begin{aligned}
(A' \setminus B) \cap (C' \setminus B) &= (A' \cap B') \cap (C' \cap B') && \text{(definition)} \\
&= (A' \cap C') \cap (B' \cap B') && \text{(reorder)} \\
&= (A \cup C)' \cap (B' \cap B') && \text{(De Morgan)} \\
&= (A \cup C)' \cap B' && \text{(involution)} \\
&= (A \cup C)' \setminus B && \text{(definition)}
\end{aligned}$$

6. Relations (6 + 8 points)

Suppose R and S are relations of type $\{1, 2, 3, 4\} \times \{\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}\}$ defined by

$$\begin{aligned}
R &:= \{\langle 1, \mathbf{a} \rangle, \langle 2, \mathbf{a} \rangle, \langle 2, \mathbf{b} \rangle, \langle 3, \mathbf{b} \rangle, \langle 3, \mathbf{c} \rangle, \langle 4, \mathbf{c} \rangle\}, \\
S &:= \{\langle 1, \mathbf{b} \rangle, \langle 2, \mathbf{d} \rangle, \langle 3, \mathbf{a} \rangle, \langle 4, \mathbf{b} \rangle, \langle 4, \mathbf{d} \rangle\}.
\end{aligned}$$

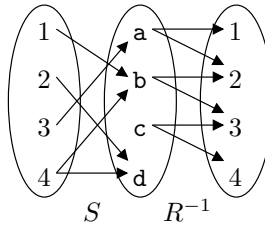
- (a) Write down the matrix representations of the relations R and S .
- (b) Draw a Venn diagram depicting the composition $R^{-1} \circ S$, and explicitly list all the elements of the composite relation $R^{-1} \circ S$ in the curly-bracket notation.

Solution:

- (a) If we label the rows in the order 1, 2, 3, 4 from top to bottom, and the columns in the order $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$ from left to right, the two matrix representations are:

$$R: \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \qquad S: \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

(b)



$$R^{-1} \circ S = \{\langle 1, 2 \rangle, \langle 1, 3 \rangle, \langle 3, 1 \rangle, \langle 3, 2 \rangle, \langle 4, 2 \rangle, \langle 4, 3 \rangle\}$$

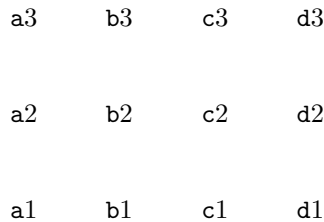
7. Ordering relation (8 + 8 points)

Suppose that in each of the sets $A := \{a, b, c, d\}$ and $B := \{1, 2, 3\}$ we have a partial ordering relation that is described by, respectively, the left or right Hasse diagram shown below:



These partial ordering relations in A and B induce a lexicographic order and a cartesian order on the product set $A \times B$.

- (a) Draw the Hasse diagram for the lexicographic order on $A \times B$ by adding arrows to the diagram you see below (where **a1** is short for $\langle a, 1 \rangle$, et cetera).

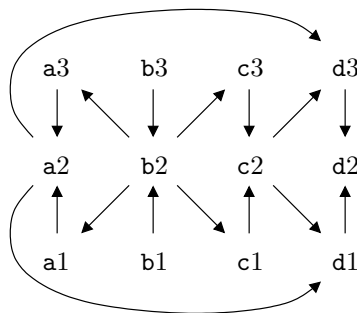


NB: you do *not* have to use the algorithm you learned to construct the Hasse diagram and you do *not* have to explain how you obtained your diagram.

- (b) Likewise, draw the Hasse diagram for the cartesian order on $A \times B$.

Solution:

- (a) Lexicographic:



- (b) Cartesian:

